

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	04 July 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Registration	USN-1	As a user, I can register for the application using my name, email, password, and location.	3	High	Frontend Developer, Backend Developer
Sprint-1	Registration	USN-2	As a user, I receive an email verification after registration.	2	High	Backend Developer
Sprint-1	Login	USN-3	As a user, I can log into the application using my email and password.	2	High	Frontend Developer, Backend Developer
Sprint-1	Dashboard	USN-4	As a user, I can view the dashboard showing current weather conditions and flood risk status.	5	High	Frontend Developer, Backend Developer
Sprint-1	User Profile	USN-5	As a user, I can update my personal profile and preferred location.	2	Medium	Frontend Developer
Sprint-1	Registration	USN-1	As a user, I can register for the application using my name, email, password, and location.	3	High	Frontend Developer, Backend Developer

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Registration	USN-2	As a user, I receive an email verification after registration.	2	High	Backend Developer
Sprint-1	Login	USN-3	As a user, I can log into the application using my email and password.	2	High	Frontend Developer, Backend Developer
Sprint-2	Weather Data Integration	USN-6	As a system, I can collect real-time weather data from external APIs.	5	High	Backend Developer
Sprint-2	River Level Monitoring	USN-7	As a system, I can collect river water level data from sensors or APIs.	5	High	Backend Developer
Sprint-2	Machine Learning Model	USN-8	As a system, I pre-process historical flood datasets before prediction.	8	High	ML Engineer
Sprint-2	Machine Learning Model	USN-9	As a system, I predict flood risk using machine learning algorithms.	8	High	ML Engineer, Backend Developer
Sprint-2	Database	USN-10	As an administrator, I can store prediction results and historical flood records.	3	Medium	Database Administrator
Sprint-3	Flood Alerts	USN-11	As a user, I receive instant flood warning notifications based on my location.	5	High	Backend Developer
Sprint-3	GIS Maps	USN-12	As a user, I can view flood-prone areas on an interactive map.	5	High	Frontend Developer, GIS Developer
Sprint-3	Reports	USN-13	As an administrator, I can generate flood prediction reports.	3	Medium	Backend Developer
Sprint-3	Emergency Contacts	USN-14	As a user, I can access emergency contact information and nearby shelters.	2	Medium	Frontend Developer

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
sprint-1	20	6 Days	29 Jun 2026	02 Jul 2026	20	04 Jul 2026
Sprint-2	20	6 Days	02 Jul 2026	03 Jul 2026	20	04 Jul 2026
Sprint-3	20	6 Days	03 Jul 2026	04 Jul 2026	20	04 Jul 2026

Velocity: Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>

<https://www.atlassian.com/agile/tutorials/burndown-charts>

Reference:

<https://www.atlassian.com/agile/project-management>

<https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software>